

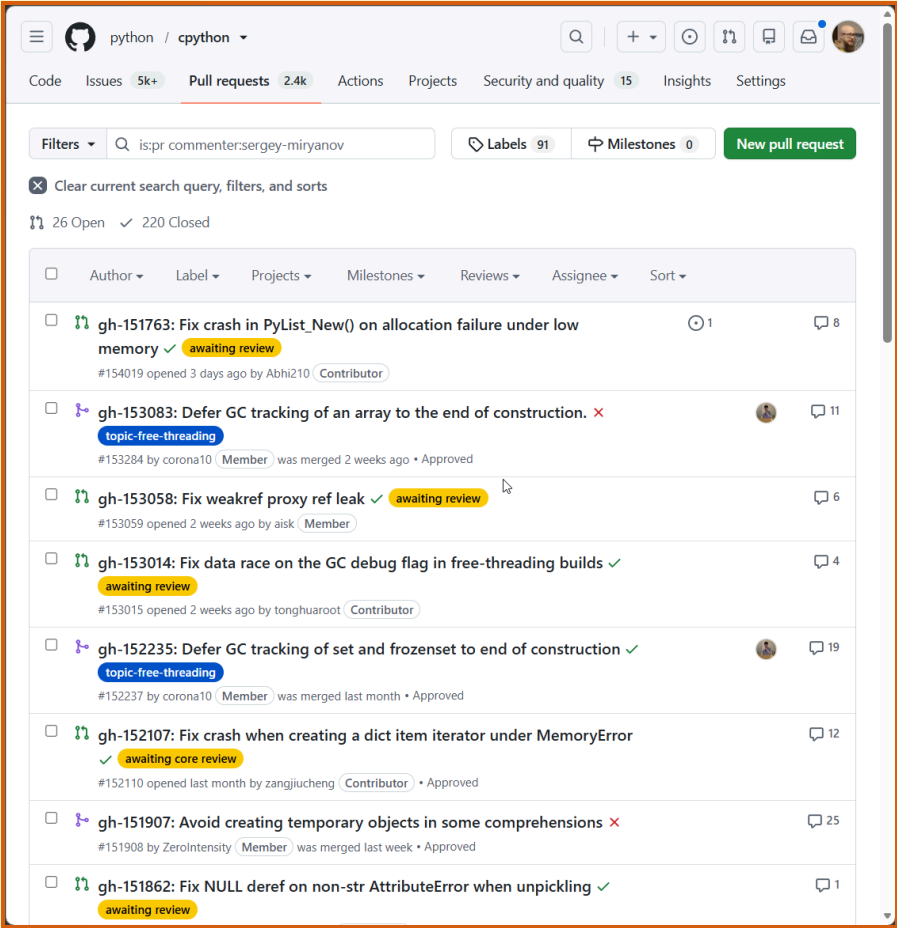
Monitoring Garbage Collector Events in Python

Yesterday, today, tomorrow?

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PyCon Russia 2026

CPYTHON CONTRIBUTOR



Python Developer's Guide

This guide is a comprehensive resource for contributing to [Python](#) – for both new and experienced contributors. It is [maintained](#) by the same community that maintains Python. We welcome your contributions!

Start with the area that best matches what you want to do. If you still have questions after reviewing the material in this guide, then the [Core Python Mentorship](#) group is available to help guide new contributors through the process.

Documentation	Code	Triage
Helping with documentation Getting started Style guide reStructuredText primer Translating Helping with the Developer's Guide Guidelines for using AI tools	Setup and building Where to get help Lifecycle of a pull request Running and writing tests Fixing "easy" issues (and beyond) Following Python's development Git bootcamp and cheat sheet Development cycle Guidelines for using AI tools	Issue tracker Triageing an issue Helping triage issues Experts index GitHub labels GitHub issues for BPO users Triage Team

We **recommend** that sections of this guide be read as needed. You can stop where you feel comfortable and begin contributing immediately without reading and understanding everything. If you do choose to skip around within the guide, be aware that some sections build on each other, so you may need to backtrack for missing concepts or terminology.

WHAT WE'LL TALK ABOUT

- Memory management
- Garbage collection (GC, Garbage Collector, Garbage Collection)
- Garbage collection events

PROBLEM

Problem ●

Tools ○

New API ○

gcmon ○

Perfetto ○

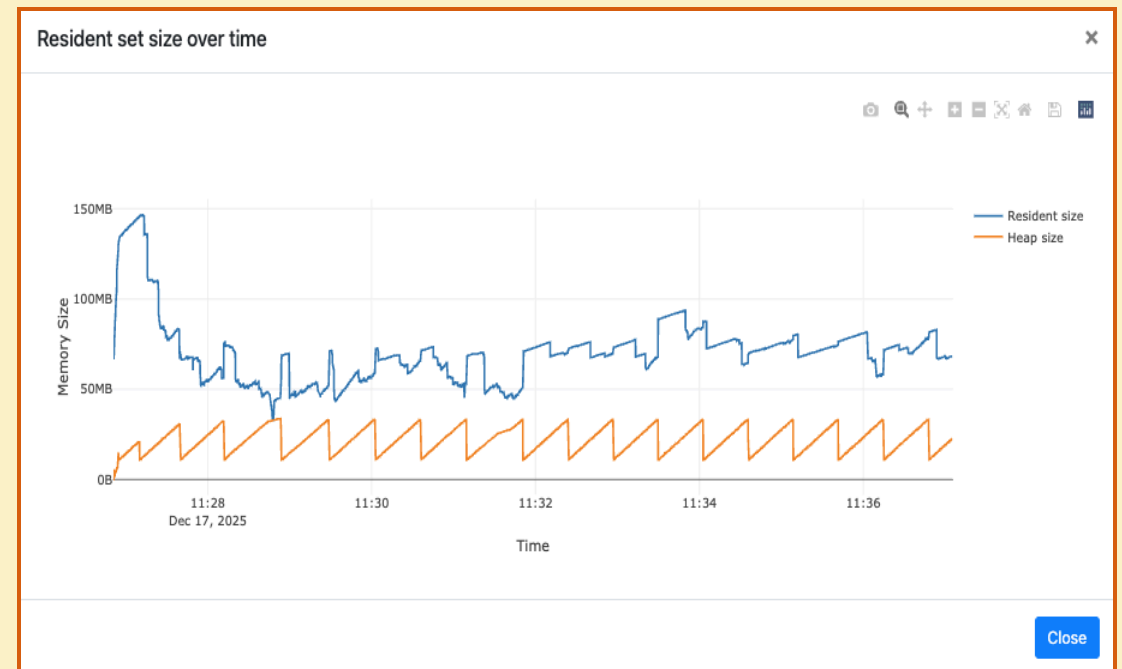
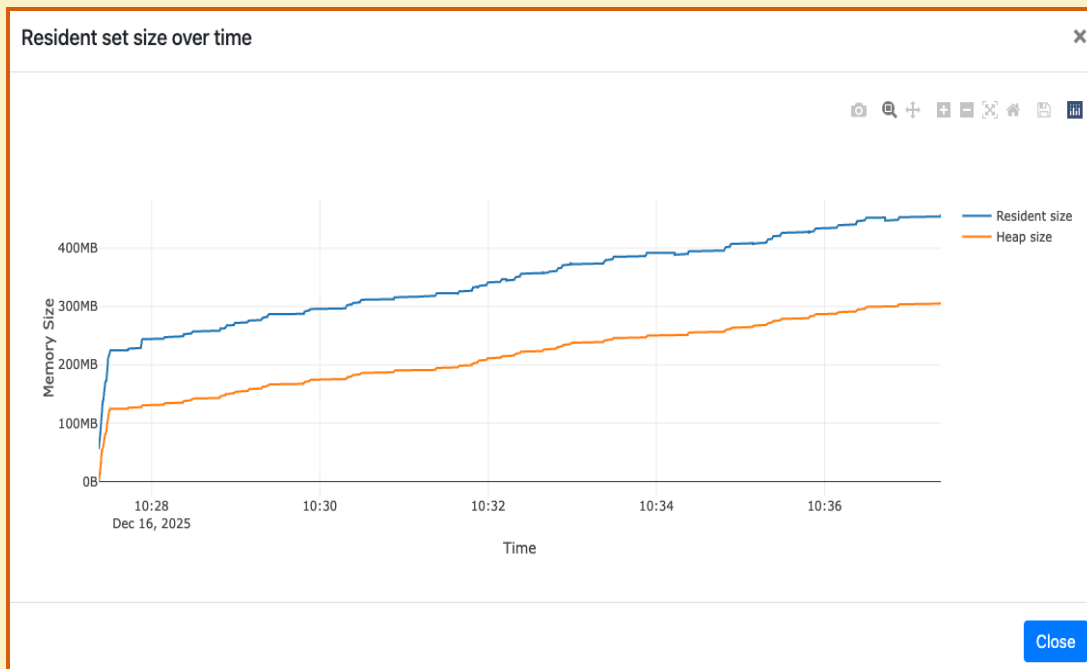
Benchmarks ○

Backports ○

<https://github.com/python/cpython/issues/142516>

3.14

3.13

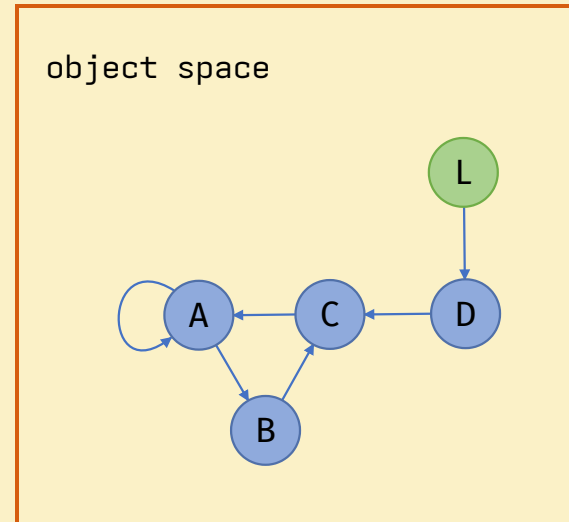
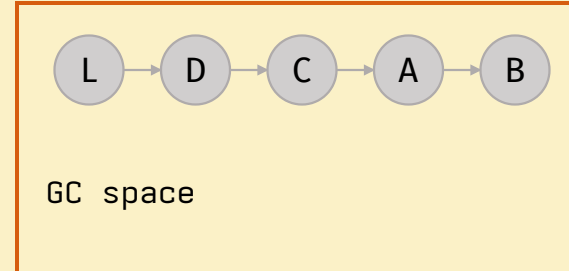


WHY DOES MEMORY USAGE GROW?

- A natural state?
- An internal leak?
- An external leak?
- A bug in the Python runtime/stdlib?
- A bug in the GC?

WHAT IS A MEMORY LEAK?

- Forgot to update the reference count
- Didn't free an unmanaged memory block
- Extended the object's lifetime
- The garbage collector didn't break cycles



gc_next
gc_prev

ob_refcnt

ob_type

...

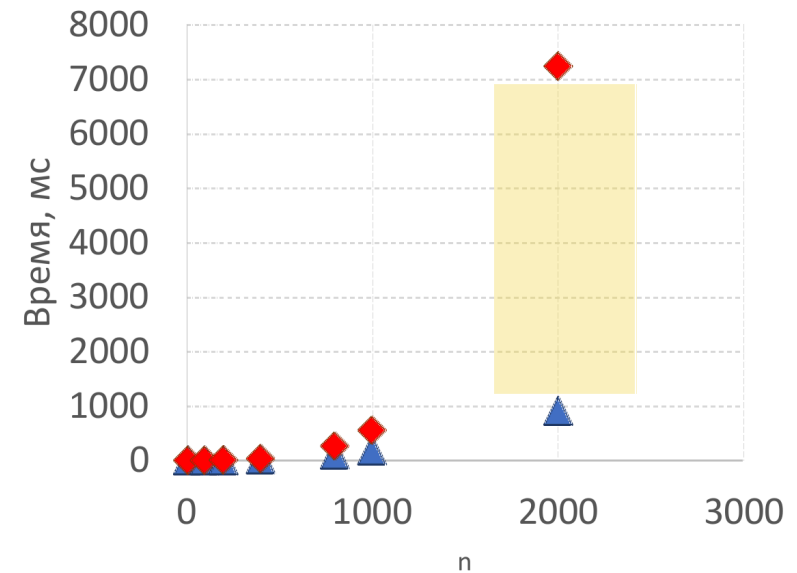
...

memory layout

WHAT IS THE GC RESPONSIBLE FOR?

- Identifies unreachable objects
- Calls finalizers
- Handles weak references
- Breaks cycles
- *Deletes objects*

```
def test(n=2000):  
    d = {}  
    for i in range(n):  
        for j in range(n):  
            d[(i,j)] = i
```



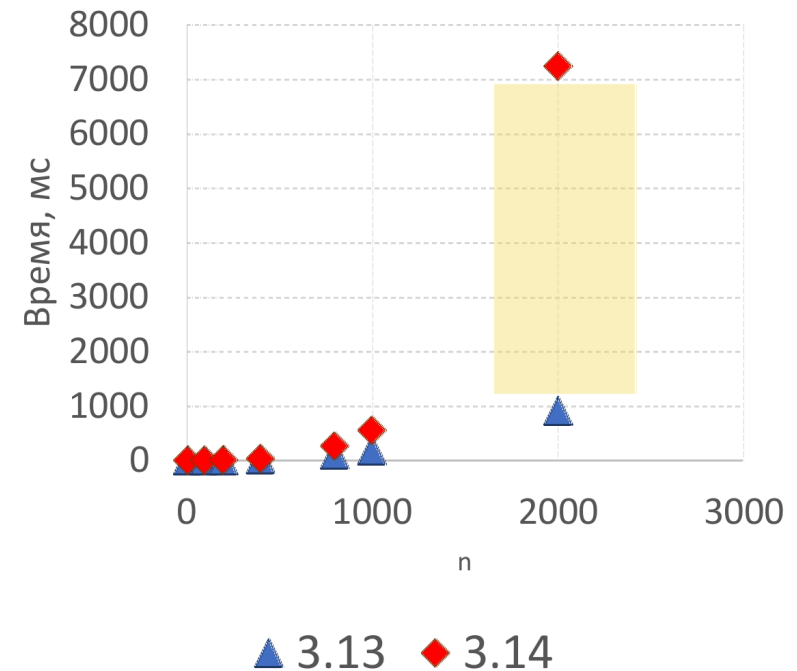
▲ 3.13 ◆ 3.14

WHAT IS THE GC RESPONSIBLE FOR?

- 3.14.0-3.14.4
 - Identifies live objects
 - Builds the increment
- Identifies unreachable objects
- Calls finalizers
- Handles weak references
- Breaks cycles
- *Deletes objects*

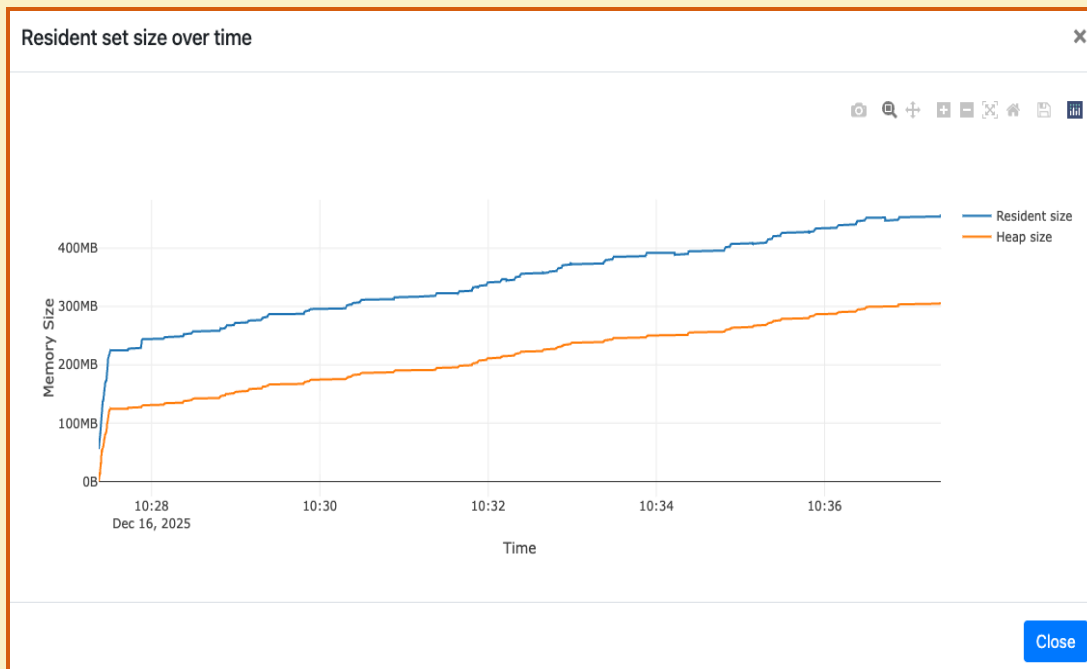


```
def test(n=2000):  
    d = {}  
    for i in range(n):  
        for j in range(n):  
            d[(i,j)] = i
```



<https://github.com/python/cpython/issues/142516>

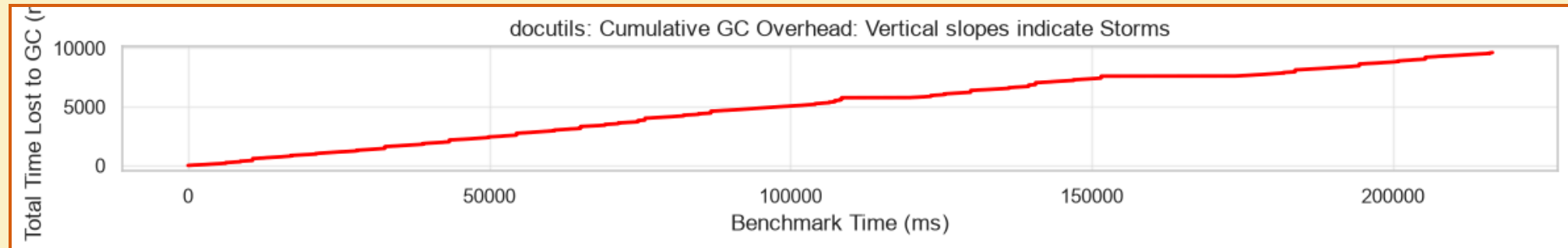
3.14



```
1 gc: collectable <Response 0x7fd8e3c4cc20>
2   → <bytes 0x561a014e9218 rc=3221225472> <b''>
3 >   → <IdentityDecoder 0x7fd8e3c4d010 rc=1> <<httpx._decoders.IdentityDecoder obje
4   → <datetime.timedelta 0x7fd8e551a490 rc=1> <datetime.timedelta(microseconds=32
5   → <int 0x561a014e71f8 rc=3221225472> <0>
6   → <Request 0x7fd8e3cf4ec0 rc=1> <<Request('HEAD', 'http://www.google.com/')>>
7 >   → <str 0x7fd8e52c9650 rc=17> <'utf-8'>
45   → <dict 0x7fd8e3c2c540 rc=1> <{'http_version': b'HTTP/1.1', 'reason_phrase': b
46   → <Headers 0x7fd8e3d6ae00 rc=1> <Headers([('content-type', 'text/html; charset
71   → <list 0x7fd8e3c2c900 rc=1> <[]>
72   → <bool 0x561a014a73c0 rc=3221225472> <True>
73   → <bool 0x561a014a73c0 rc=3221225472> <True>
74   → <NoneType 0x561a014be300 rc=3221225472> <None>
75   → <int 0x561a014e8af8 rc=3221225472> <200>
76   → <BoundAsyncStream 0x7fd8e3c4cec0 rc=1> <<httpx._client.BoundAsyncStream obje
77   → <Response 0x7fd8e3c4cc20 rc=1> <<Response [200 OK]>>
78   → <bytes 0x561a014e9218 rc=3221225472> <b''>
79 >   → <IdentityDecoder 0x7fd8e3c4d010 rc=1> <<httpx._decoders.IdentityDecoder
81   → <datetime.timedelta 0x7fd8e551a490 rc=1> <datetime.timedelta(microsecond
82   → <int 0x561a014e71f8 rc=3221225472> <0>
83 >   → <Request 0x7fd8e3cf4ec0 rc=1> <<Request('HEAD', 'http://www.google.com/'
111   → <str 0x7fd8e52c9650 rc=17> <'utf-8'>
112   → <dict 0x7fd8e3c2c540 rc=1> <{'http_version': b'HTTP/1.1', 'reason_phrase
113 >   → <Headers 0x7fd8e3d6ae00 rc=1> <Headers([('content-type', 'text/html; cha
127   → <list 0x7fd8e3c2c900 rc=1> <[]>
128   → <bool 0x561a014a73c0 rc=3221225472> <True>
129   → <bool 0x561a014a73c0 rc=3221225472> <True>
130   → <NoneType 0x561a014be300 rc=3221225472> <None>
131   → <int 0x561a014e8af8 rc=3221225472> <200>
132 >   → <BoundAsyncStream 0x7fd8e3c4cec0 rc=1> <<httpx._client.BoundAsyncStream
174   → <type 0x561a239058b0 rc=30> <<class 'httpx.Response'>>
175   → <float 0x7fd8e3cdd3b0 rc=1> <39037.104077293>
176 >   → <AsyncResponseStream 0x7fd8e3cf6120 rc=1> <<httpx._transports.default.Asyn
250   → <type 0x561a23924160 rc=15> <<class 'httpx._client.BoundAsyncStream'>>
251   → <type 0x561a239058b0 rc=30> <<class 'httpx.Response'>>
```

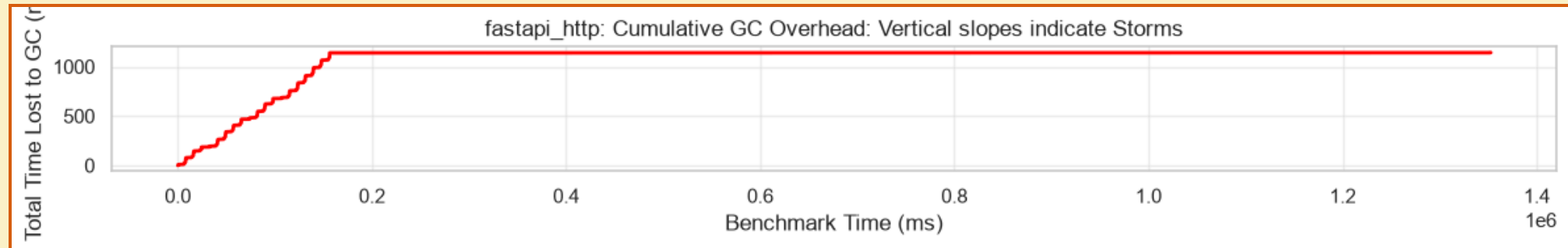
GC STORM

- Long/frequent pauses
- Increased memory/CPU usage
- Frequent full garbage collection



GC STORM

- Long/frequent pauses
- Increased memory/CPU usage
- Frequent full garbage collection



SUMMARY

- Causes of memory-usage growth
- What counts as a leak
- How reference counting works
- How the garbage collector works
- How the garbage collector affects the program
- What GC storm/thrashing is

TOOLS

Problem ●

Tools ●

New API ○

gcmon ○

Perfetto ○

Benchmarks ○

Backports ○

TOOLS

- **In-process**

- `tracemalloc`
- `pympler`
- APM (ddtrace, dynatrace, ...)

- **Out-of-process**

- `memray`
- `austin`, `tachyon`
- `pyroscope`, `perforator`

- CPU
- Allocations
- Snapshots
- Cycles
- Lifetime
- Leaks
- Number of GC pauses
- Number of collected objects

THE GC MODULE

- **What's wrong with** `gc.callbacks`
 - Intrusive
 - C->Python interop
 - Subinterpreters
 - Heisenberg effect
- **What's wrong with** `gc.get_stats`
 - Intrusive
 - When
 - Subinterpreters

NEW API

Problem ●

Tools ●

New API ●

gcmon ○

Perfetto ○

Benchmarks ○

Backports ○

GCMONITOR

- `_remote_debugging.get_gc_stats`
- `_remote_debugging.GCMonitor`
 - `init(pid)`
 - `get_gc_stats`

- External process
- Standard metrics
 - Collections
 - Collected
 - Uncollectable
 - *Candidates*
 - *Duration*
- *Number of live objects*
- Pause start and end time
- Subinterpreter support

GCMONITOR

```
struct pyruntimestate {
    /* This field must be first to facilitate locating it by out of process
     * debuggers. Out of process debuggers will use the offsets contained in
     * this field to be able to locate other fields in several interpreter
     * structures in a way that doesn't require them to know the exact layout
     * of those structures.
     *
     * IMPORTANT:
     * This struct is **NOT** backwards compatible between minor version of the
     * interpreter and the members, order of members and size can change
     * between minor versions. This struct is only guaranteed to be stable
     * between patch versions for a given minor version of the interpreter.
     */
    _Py_DebugOffsets debug_offsets;
    // ...
    struct pyinterpreters {
        PyMutex mutex;
        /* The linked list of interpreters, newest first. */
        PyInterpreterState *head;
        PyInterpreterState *main;
        int64_t next_id;
    } interpreters;
    // ...
};
```

```
typedef struct _Py_DebugOffsets {
    char cookie[8] _Py_NONSTRING;
    uint64_t version;
    uint64_t free_threaded;
    struct _runtime_state {
        uint64_t size;
        uint64_t finalizing;
        uint64_t interpreters_head;
    } runtime_state;
    struct _interpreter_state {
        uint64_t size;
        uint64_t id;
        uint64_t next;
        uint64_t gc;
        // ...
    } interpreter_state;
    struct _gc {
        uint64_t size;
        uint64_t collecting;
        uint64_t frame;
        uint64_t generation_stats_size;
        uint64_t generation_stats;
    } gc;
} _Py_DebugOffsets;
```

GCMONITOR

```
typedef struct _Py_DebugOffsets {
    char cookie[8] _Py_NONSTRING;
    uint64_t version;
    uint64_t free_threaded;
    uint64_t finalizing;
    struct _runtime_state {
        uint64_t size;
        uint64_t finalizing;
        uint64_t interpreters_head;
    } runtime_state;
    struct _interpreter_state {
        uint64_t size;
        uint64_t id;
        uint64_t next;
        uint64_t gc;
    } interpreter_state;
    struct _gc {
        uint64_t size;
        uint64_t collecting;
        uint64_t frame;
        uint64_t generation_stats_size;
        uint64_t generation_stats;
    } gc;
} _Py_DebugOffsets;
```

```
gscope tui —PID 24280 —Frame 122 @ 12.6s —Rate 180ms —Poll 128.6µs
| _Py_DebugOffsets (embedded in _PyRuntime) @ 0x7ffbba7f1808 (size: 880 bytes)

Fields:
Py_DebugOffsets
  -- 0x0000 cookie[8] "xdebugpy"
  -- 0x0008 version 0x030f00b1
  -- 0x0010 free_threaded 0
  -- runtime_state
    -- 0x0018 size 345996
    -- 0x0020 finalizing 984
    -- 0x0028 interpreters_head 928
  -- interpreter_state
    -- 0x0030 size 226016
    -- 0x0038 id 7272
    -- 0x0040 next 7264
    -- 0x0048 threads_head 7328
    -- 0x0050 threads_main 7344
    -- 0x0058 gc 7392
    -- 0x02e8 size 176
    -- 0x02f0 collecting 112
    -- 0x02f8 frame 120
    -- 0x0300 generation_stats_size 1112
    -- 0x0308 generation_stats 104
    -- item_size 64
    -- young_entries (11) 11 x 64 = 704 bytes
    -- entry
      -- ts_start +8
      -- ts_stop +8
      -- collections +16
      -- collected +24
      -- uncollectable +32
      -- candidates +40
      -- duration +48
      -- heap_size +56
    -- index0 +784
    -- old0_entries (3) 3 x 64 bytes
    -- index1 +984
    -- old1_entries (3) 3 x 64 bytes
    -- index2 +1184

Hex Dump (DebugOffsets, 0x0000-0x036f, 880 bytes):
00000000 72 64 03 02 75 07 72 72 b1 00 0f 03 00 00 00 00 xdebugpy.....
00000010 00 00 00 00 00 00 00 00 00 44 05 00 00 00 00 00 .....D.....
00000020 00 03 00 00 00 00 00 00 00 03 00 00 00 00 00 .....h.....
00000030 e0 72 03 00 00 00 00 00 68 1c 00 00 00 00 00 .....f.....
00000040 00 1c 00 00 00 00 00 00 a0 1c 00 00 00 00 00 .....r.....
00000050 00 1c 00 00 00 00 00 00 a0 1c 00 00 00 00 00 .....r.....
00000060 a0 1d 00 00 00 00 00 00 90 1d 00 00 00 00 00 .....r.....
00000070 90 1d 00 00 00 00 00 00 10 00 00 00 00 00 .....r.....
00000080 20 1e 00 00 00 00 00 00 00 00 00 00 00 00 .....r.....
00000090 30 1e 00 00 00 00 00 00 28 1e 00 00 00 00 00 .....r.....
000000a0 50 1e 00 00 00 00 00 00 00 00 00 00 00 00 .....r.....
000000b0 68 03 00 00 00 00 00 00 00 00 00 00 00 00 .....r.....
000000c0 00 00 00 00 00 00 00 00 10 00 00 00 00 00 .....r.....
000000d0 48 00 00 00 00 00 00 00 50 00 00 00 00 00 .....r.....
000000e0 58 00 00 00 00 00 00 00 a8 00 00 00 00 00 .....r.....
000000f0 ac 00 00 00 00 00 00 00 f0 00 00 00 00 00 .....r.....
00000100 50 00 00 00 00 00 00 00 4a 00 00 00 00 00 .....r.....
00000110 28 00 00 00 00 00 00 00 80 00 00 00 00 00 .....r.....
00000120 10 01 00 00 00 00 00 00 00 00 00 00 00 00 .....r.....
00000130 58 00 00 00 00 00 00 00 08 00 00 00 00 00 .....r.....
00000140 00 00 00 00 00 00 00 00 38 00 00 00 00 00 .....r.....
00000150 50 00 00 00 00 00 00 00 4a 00 00 00 00 00 .....r.....
00000160 40 00 00 00 00 00 00 00 00 00 00 00 00 00 .....r.....
00000170 d8 00 00 00 00 00 00 00 70 00 00 00 00 00 .....r.....
00000180 78 00 00 00 00 00 00 00 80 00 00 00 00 00 .....r.....
00000190 88 00 00 00 00 00 00 00 44 00 00 00 00 00 .....r.....
000001a0 24 00 00 00 00 00 00 00 60 00 00 00 00 00 .....r.....
000001b0 68 00 00 00 00 00 00 00 00 00 00 00 00 00 .....r.....
000001c0 00 00 00 00 00 00 00 00 10 00 00 00 00 00 .....r.....
000001d0 08 00 00 00 00 00 00 00 a8 01 00 00 00 00 .....r.....
000001e0 18 00 00 00 00 00 00 00 58 00 00 00 00 00 .....r.....
000001f0 40 00 00 00 00 00 00 00 20 00 00 00 00 00 .....r.....
00000200 20 01 00 00 00 00 00 00 b0 03 00 00 00 00 .....r.....
00000210 70 03 00 00 00 00 00 00 20 00 00 00 00 00 .....r.....
00000220 20 00 00 00 00 00 00 00 10 00 00 00 00 00 .....r.....
00000230 20 00 00 00 00 00 00 00 18 00 00 00 00 00 .....r.....
00000240 10 00 00 00 00 00 00 00 c8 00 00 00 00 00 .....r.....
00000250 10 00 00 00 00 00 00 00 28 00 00 00 00 00 .....r.....
00000260 20 00 00 00 00 00 00 00 30 00 00 00 00 00 .....r.....
00000270 20 00 00 00 00 00 00 00 28 00 00 00 00 00 .....r.....
00000280 18 00 00 00 00 00 00 00 10 00 00 00 00 00 .....r.....
00000290 20 00 00 00 00 00 00 00 10 00 00 00 00 00 .....r.....
000002a0 10 00 00 00 00 00 00 00 28 00 00 00 00 00 .....r.....
000002b0 10 00 00 00 00 00 00 00 20 00 00 00 00 00 .....r.....
000002c0 40 00 00 00 00 00 00 00 20 00 00 00 00 00 .....r.....
000002d0 10 00 00 00 00 00 00 00 28 00 00 00 00 00 .....r.....
000002e0 38 00 00 00 00 00 00 00 38 00 00 00 00 00 .....r.....
000002f0 70 00 00 00 00 00 00 00 78 00 00 00 00 00 .....r.....
00000300 58 04 00 00 00 00 00 00 48 00 00 00 00 00 .....r.....
00000310 a0 00 00 00 00 00 00 00 18 00 00 00 00 00 .....r.....
00000320 48 00 00 00 00 00 00 00 43 00 00 00 00 00 .....r.....
00000330 00 00 00 00 00 00 00 00 08 00 00 00 00 00 .....r.....
00000340 18 00 00 00 00 00 00 00 48 01 00 00 00 00 .....r.....
00000350 a8 1e 00 00 00 00 00 00 80 00 00 00 00 00 .....r.....
00000360 04 00 00 00 00 00 00 00 00 02 00 00 00 00 .....r.....

PyInterpreterState @ 0x7ffbba80e128 (struct: 226016 bytes)

Key offset values stored in _Py_DebugOffsets:
  interpreter_state.id 7272
  interpreter_state.next 7264
  interpreter_state.threads_head 7328

GC struct (176 bytes) hex dump:
00000000 01 00 00 00 00 00 00 00 00 4a 8a d5 98 01 00 00 .....J.....
00000010 40 0f d7 d5 98 01 00 00 d0 07 00 00 00 03 00 00 .....@.....
00000020 20 fe 00 ba fb 7f 00 00 20 fe 00 ba fb 7f 00 00 .....C.....

[q] quit [s] save [p] pick pid [g] gc-view [t] tree [h] hex [c] collapse [d] dbg/rtr/rw [e] glitch-off [f/j/k] entry 1/17 [PageUp/Down] go
```

GC STATISTICS

```
struct gc_generation_stats {
    PyTime_t ts_start;
    PyTime_t ts_stop;
    Py_ssize_t collections;
    Py_ssize_t collected;
    Py_ssize_t uncollectable;
    Py_ssize_t candidates;
    double duration;
    Py_ssize_t heap_size;
};

struct gc_young_stats_buffer {
    struct gc_generation_stats items[11];
    int8_t index;
};

struct gc_old_stats_buffer {
    struct gc_generation_stats items[3];
    int8_t index;
};

struct gc_stats {
    struct gc_young_stats_buffer young;
    struct gc_old_stats_buffer old[2];
};
```

gcscope tui —PID 24288 —frame 766 @ 83.1s —Rate 100ms —Poll 76.1µs

GC Stats Buffer View ([g] back to full layout)

Buffer @ 0x198d51de470 (size: 1112 bytes, entry: 64 bytes)

Gen 0 (Young) - 11 entries (rate = 3.33/s, avg coll = 6.880ms)

Gen 1 (Middle) - 3 entries (rate = 0.32/s, avg coll = 5.632ms)

Gen 2 (Oldest) - 3 entries (rate = 0.8/s, avg coll = 10.781ms)

Legend: ring = per-generation ring index (18, points at the active entry)

gen	idx	collections	collected	heap	duration(s)
0	0	594869	1186154611	11.8K	3672.586
0	1	594870	1186156605	11.8K	3672.393
0	2	594871	1186158599	11.8K	3672.399
0	3	594872	1186160593	11.8K	3672.405
0	4	594882	1186148653	11.8K	3672.344
0	5	594863	1186142647	11.8K	3672.351
0	6	594864	1186144641	11.8K	3672.357
0	7	594865	1186146635	11.8K	3672.363
0	8	594866	1186148629	11.8K	3672.369
0	9	594867	1186150623	11.8K	3672.375
0	10	594868	1186152617	11.8K	3672.381
1	0	54078	107831532	11.8K	334.103
1	1	54076	10827544	11.8K	334.092
1	2	54077	107829538	11.8K	334.098
2	0	156	309070	11.8K	1.273
2	1	154	305882	11.8K	1.252
2	2	155	307876	11.8K	1.260

Entry #1 (gen 0, entry 0) @ 0x198d51de470

ts_start	0	+0	266,436,333,253,680
ts_stop	0	+8	266,436,339,072,680
collections	0	+16	594869
collected	0	+24	1186154611
uncollectable	0	+32	0
candidates	0	+40	1191522914
duration	0	+48	3672.366371
heap_size	0	+56	11778

[q] quit [s] save [p] pick pid [u] go-view [t] tree [h] hex [c] collapse [d] dbg/ht [r/r] raw [g] glitch/off [t/t]k entry 1/17 [page/page] scroll

PAUSES

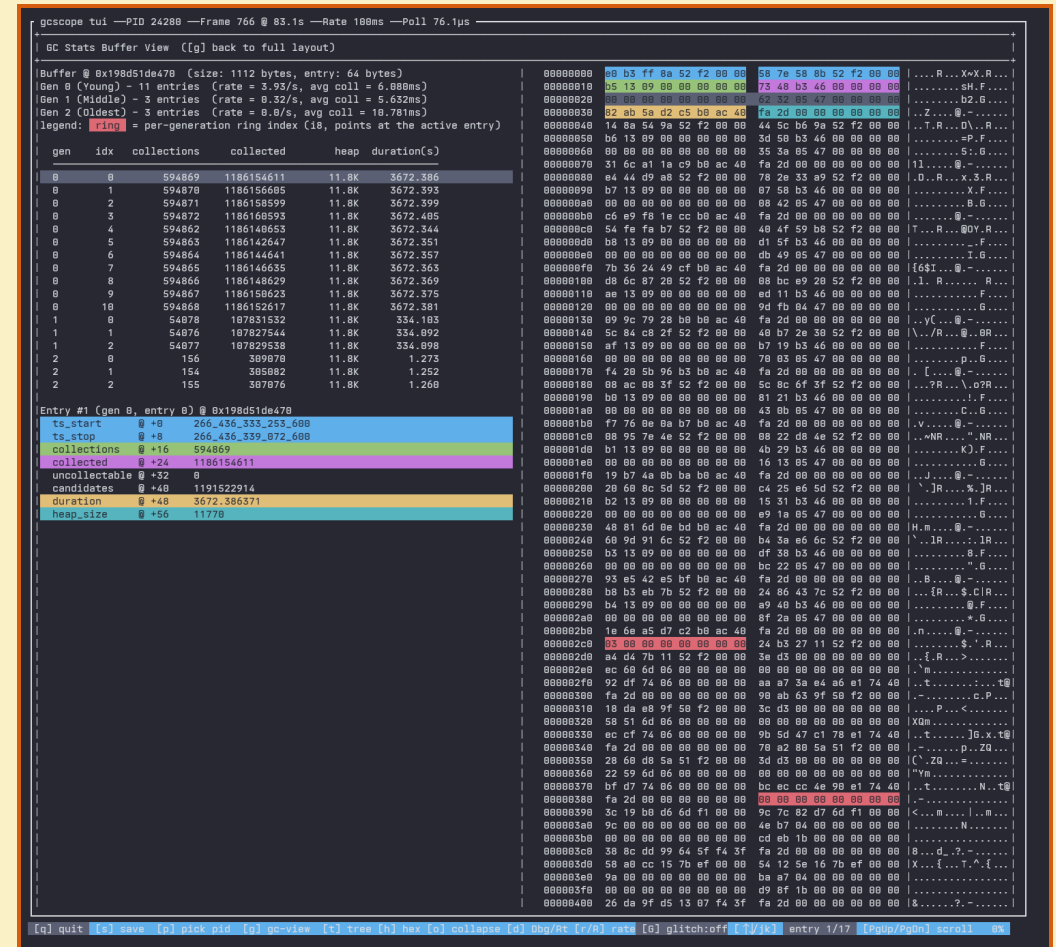
```
struct _PyInterpreterFrame {
    _PyStackRef f_executable;
    struct _PyInterpreterFrame *previous;
    _PyStackRef f_funcobj;
    PyObject *f_globals;
    PyObject *f_builtins;
    PyObject *f_locals;
    PyFrameObject *frame_obj;
    _Py_CODEUNIT *instr_ptr;
    _PyStackRef *stackpointer;
    uint16_t return_offset;
    char owner;
    uint8_t visited;
    /* Locals and stack */
    _PyStackRef localsplus[1];
};
```

previous

frame

previous

frame



PITFALLS

- Non-Python processes
- Child processes
- Race condition
 - Incomplete initialization
 - Partial read
- Elevated privileges
- GC storm/thrashing

SUMMARY

- Existing tools answer different questions
- Using the GC module distorts measurements
- The new API lets you look inside the GC without distortion
- The new API requires no pauses
- But there are pitfalls to keep in mind

GCMON

Problem ●

Tools ●

New API ●

gcmmon ●

Perfetto ○

Benchmarks ○

Backports ○

GCMON

- **Process management**

- Child processes
- Lifetime
- Metadata
- RSS

- **Extended metrics**

- **Export**

- Various formats

- `pyperf` support

- Labels

- **Overhead**

- Rate=100ms -> 1%
- Rate=10ms -> 3-4%

- What about gcscope?

- Perfetto Binary Format

- Chrome Trace Format

- JSONL/stdout

- OpenTelemetry?

GCMON

- **Standard metrics**
 - Collections
 - Collected
 - Uncollectable
 - *Candidates*
 - *Duration*
- *Number of live objects*
- Pause start and end time
- RSS

GCMON

- **Debug data**
 - Mark Alive
 - Fill Increment
 - Deduce Unreachable
 - Handle Weakrefs Callbacks
 - Finalize Garbage
 - Handle Resurrected
 - Clear Weakrefs
 - Delete Garbage

PERFETTO

Problem ●

Tools ●

New API ●

gcmon ●

Perfetto ●

Benchmarks ○

Backports ○

CURRENT TRACE

pyperformance.pftrace (1 MB)

Timeline

Overview

Data Explorer

Query (SQL)

Metrics

Download

NEW TRACE

Open trace file

Open multiple trace files

Record new trace

Open Android example

Open Chrome example

SETTINGS

Settings

Flags

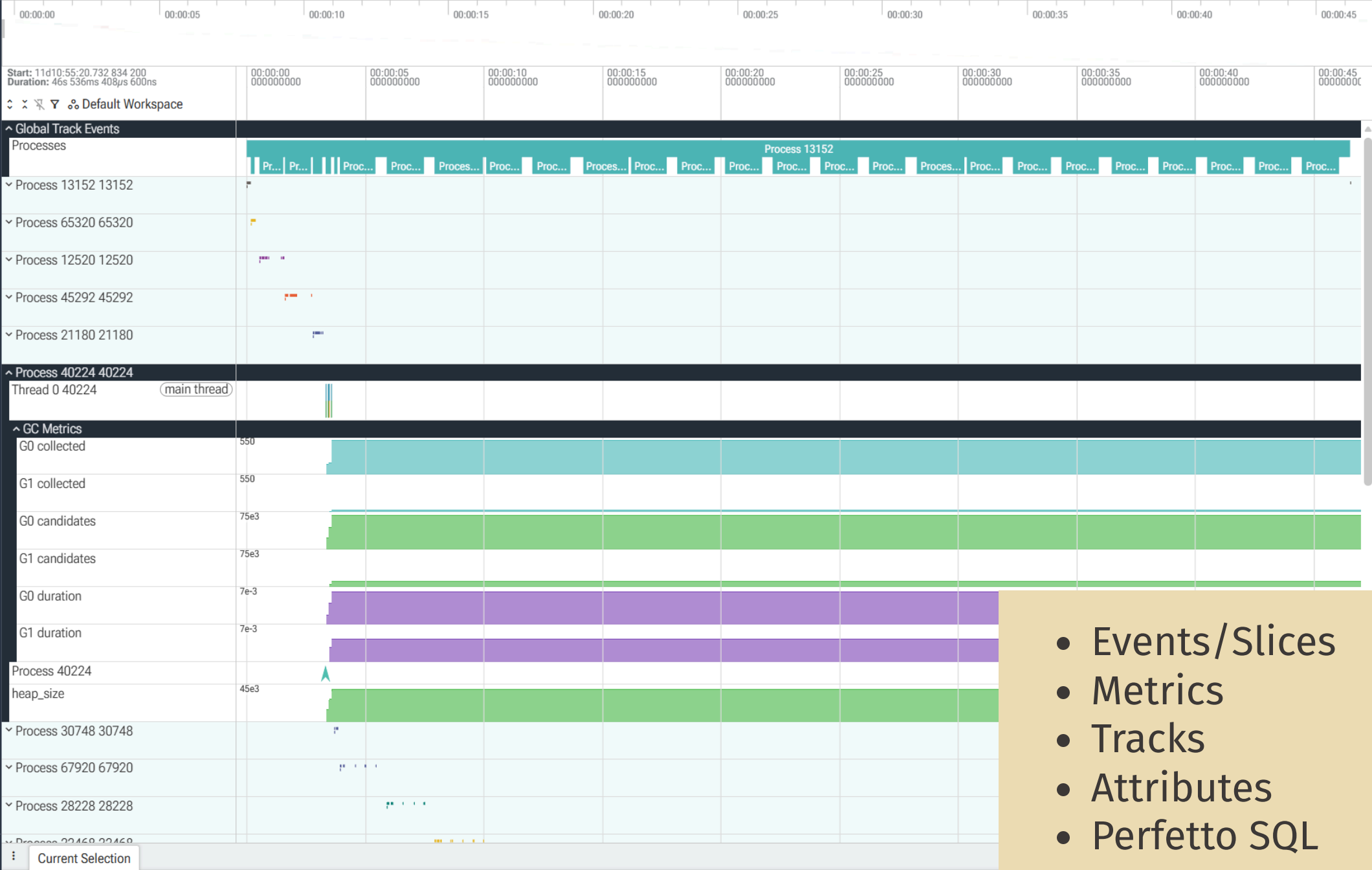
Plugins

SUPPORT

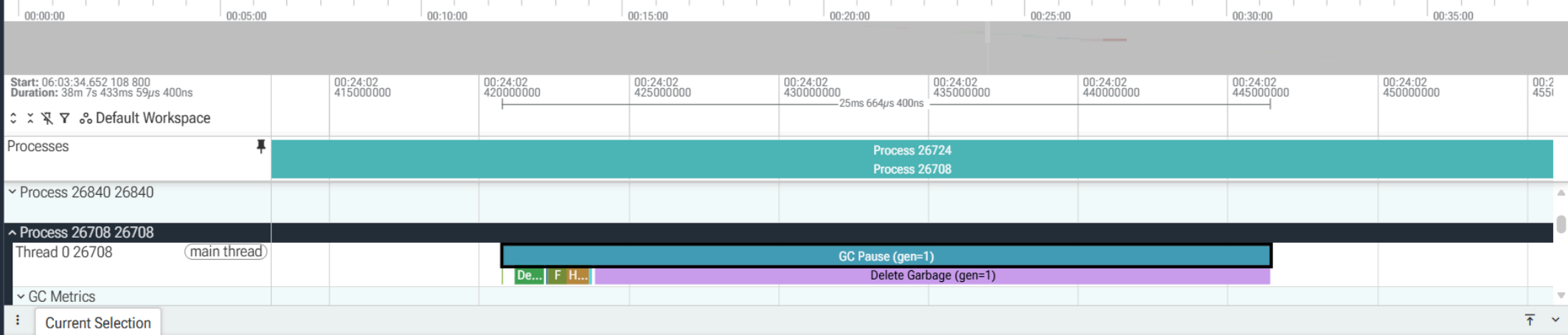
Documentation & Bugs

CONVERT TRACE

Convert to other formats



- Events/Slices
- Metrics
- Tracks
- Attributes
- Perfetto SQL



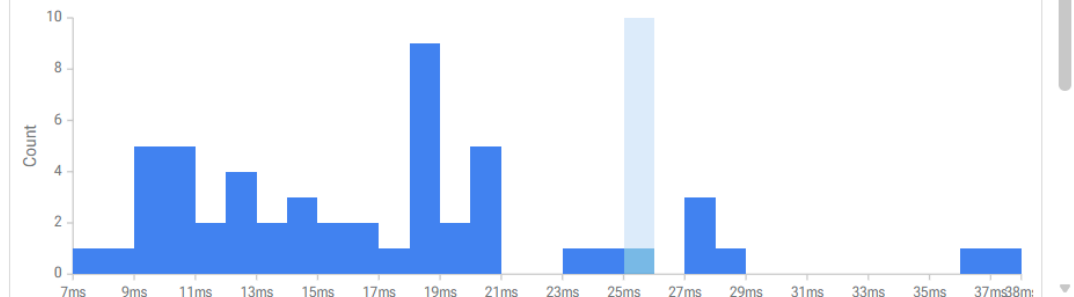
Slice GC Pause (gen=1)

Category [gc.pause\(gen=1\)](#)
 Start time [00:24:02.420 760 100](#)
 Duration [25ms 664μs 400ns](#)
 Thread [Thread 0 \[26708\]](#)
 Process [Process 26708 \[26708\]](#)
 SQL ID [slice\[51794\]](#)

generation 1
 iid 0
 collections 22
 heap_size 147289
 collected 345488
 uncollectable 0
 candidates 456896
 increment_size 8019
 alive_size 37319
 finalized_garbage_count 31408
 deleted_garbage_count 31408
 clear_weakrefs_count 0

Histogram for 'GC Pause (gen=1)'

Scope: This track



- Process minimap
- Attributes
- Garbage collection phases

CURRENT TRACE

cyclotron.pftrace (37 MB)

Timeline

Overview

Data Explorer

Query (SQL)

Metrics

Download

NEW TRACE

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Record new trace

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Open Chrome example

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Flags

Plugins

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Documentation & Bugs

CONVERT TRACE

Convert to other formats

<> Query 1 +

Run Query

or press Ctrl ↵

Format

History

Tables

Query history (50 queries)

```
1 SELECT
2   name,
3   COUNT(dur) AS count,
4   ROUND(SUM(dur) / 1e6, 2) as dur_ms,
5   ROUND(AVG(dur) / 1e6, 4) AS avg_ms,
6   -- Calculate P50, P90, and P99 in milliseconds
7   ROUND(PERCENTILE(dur, 50) / 1e6, 4) AS P50_dur_ms,
8   ROUND(PERCENTILE(dur, 90) / 1e6, 4) AS P90_dur_ms,
9   ROUND(PERCENTILE(dur, 95) / 1e6, 4) AS P95_dur_ms,
10  ROUND(PERCENTILE(dur, 99) / 1e6, 4) AS P99_dur_ms
11 FROM slice
12 WHERE category IS NOT NULL
13 GROUP BY name
14 ORDER BY IIF(parent_id IS NULL, 0, 1), name
```

Returned 24 rows in 125 ms

Add debug track Export

name	count	dur_ms	avg_ms	P50_dur_ms	P90_dur_ms	P95_dur_ms	F
GC Pause (gen=0)	33746	6741.95	0.1998	0.0959	0.3035	0.4199	
GC Pause (gen=1)	1980	11499.15	5.8077	2.3585	5.9265	17.9474	
GC Pause (gen=2)	49	5074.84	103.5682	12.4377	171.8633	296.0494	
Clear Weakrefs (gen=0)	19788	72.07	0.0036	0.0029	0.0069	0.0079	
Clear Weakrefs (gen=1)	1954	91.29	0.0467	0.0472	0.0887	0.1005	
Clear Weakrefs (gen=2)	49	53.57	1.0932	0.1835	1.7276	2.2792	
Deduce Unreachable (gen=0)	33746	1964.59	0.0582	0.0467	0.0918	0.1089	
Deduce Unreachable (gen=1)	1980	1312.16	0.6627	0.5931	1.1439	1.3738	
Deduce Unreachable (gen=2)	49	167.34	3.4152	2.6387	9.3456	9.656	
Delete Garbage (gen=0)	21046	3249.07	0.1544	0.0417	0.149	0.2601	
Delete Garbage (gen=1)	1952	7705.45	3.9475	0.6006	3.0049	14.5245	
Delete Garbage (gen=2)	49	4229.49	86.316	2.9848	81.8673	262.4667	
Fill increment (gen=1)	1978	97.9	0.0495	0.0398	0.0949	0.1161	
Finalize Garbage (gen=0)	23502	490.09	0.0209	0.0059	0.0602	0.0734	
Finalize Garbage (gen=1)	1954	396.85	0.2031	0.1108	0.586	0.6976	

```
SELECT
name,
COUNT(dur) AS count,
ROUND(SUM(dur) / 1e6, 2) as dur_ms,
ROUND(AVG(dur) / 1e6, 4) AS avg_ms,
-- Calculate P50, P90, and P99 in
milliseconds
ROUND(PERCENTILE(dur, 50) / 1e6, 4) AS

SELECT
name,
COUNT(dur) AS count,
ROUND(SUM(dur) / 1e6, 2) as dur_ms,
ROUND(AVG(dur) / 1e6, 4) AS avg_ms,
-- Calculate P50, P90, and P99 in
milliseconds
ROUND(PERCENTILE(dur, 50) / 1e6, 4) AS

SELECT
name,
COUNT(dur) AS count,
ROUND(SUM(dur) / 1e6, 2) as dur_ms,
ROUND(AVG(dur) / 1e6, 4) AS avg_ms,
-- Calculate P50, P90, and P99 in
milliseconds
ROUND(PERCENTILE(dur, 50) / 1e6, 4) AS

SELECT
name,
COUNT(dur) AS count,
ROUND(SUM(dur) / 1e6, 2) as dur_ms,
ROUND(AVG(dur) / 1e6, 4) AS avg_ms,
-- Calculate P50, P90, and P99 in
milliseconds
ROUND(PERCENTILE(dur, 50) / 1e6, 4) AS

SELECT DISTINCT p.upid, p.pid , s.ts, s.dur
/ 1e6 as dur_ms, a.string_value as cmdline
FROM slice s
JOIN process p ON s.name = 'Process ' ||
```

- Working with data
- Derived metrics
- SQLite+

CURRENT TRACE

cyclotron.pftrace (37 MB)

Timeline

Overview

Data Explorer

Query (SQL)

Metrics

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NEW TRACE

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SETTINGS

Settings

Flags

Plugins

SUPPORT

Documentation & Bugs

CONVERT TRACE

Convert to other formats

<> Query 1 +

Run Query

or press **Ctrl** **↵**

Format

History

Tables

Query history (50 queries)

```
1 SELECT
2   p.ts,
3   p.dur,
4   p.name
5 FROM (SELECT * FROM slice WHERE name LIKE 'GC Pause%') p
6 INNER JOIN (SELECT * FROM slice WHERE name LIKE 'Delete Garbage%') g on p.id = g.parent_id
7 WHERE
8   (100.0 * g.dur/p.dur) > 25 AND p.dur/1e6 > 10
9
```

Returned 172 rows in 29 ms

Add debug track

Export

ts	dur	name
23085429684900	156325100	GC Pause (gen=2)
23155364252900	29611400	GC Pause (gen=2)
23159019828500	26566600	GC Pause (gen=2)
23186002205700	58527500	GC Pause (gen=2)
23255273834200	24365500	GC Pause (gen=1)
23256146116000	18237700	GC Pause (gen=1)
23256648973400	17942600	GC Pause (gen=1)
23257072868900	25664400	GC Pause (gen=1)
23257259102500	19320500	GC Pause (gen=1)
23261802915900	27232000	GC Pause (gen=1)
23262924635500	20695100	GC Pause (gen=1)
23264109573900	18039000	GC Pause (gen=1)
23267426957400	23078300	GC Pause (gen=1)
23268197362800	10299800	GC Pause (gen=1)
23269575030400	12236200	GC Pause (gen=1)
23270068817400	18046900	GC Pause (gen=1)

```
SELECT
  p.ts,
  p.dur,
  p.name
FROM (SELECT * FROM slice WHERE name LIKE
'GC Pause%') p
INNER JOIN (SELECT * FROM slice WHERE name
LIKE 'Delete Garbage%') g on p.id =

SELECT
  p.ts,
  p.dur,
  p.name
FROM (SELECT * FROM slice WHERE name LIKE
'GC Pause%') p
INNER JOIN (SELECT * FROM slice WHERE name
LIKE 'Delete Garbage%') g on p.id =

SELECT
  p.ts,
  p.dur,
  p.name
FROM (SELECT * FROM slice WHERE name LIKE
'GC Pause%') p
INNER JOIN (SELECT * FROM slice WHERE name
LIKE 'Delete Garbage%') g on p.id =

-- SELECT
-- name,
--   COUNT(dur) AS count,
--   ROUND(SUM(dur) / 1e6, 2) as dur_ms,
--   ROUND(AVG(dur) /1e6, 4) AS avg_ms,
--   -- Calculate P50, P90, and P99 in
--   milliseconds
--   ROUND(PERCENTILE(dur, 50) / 1e6, 4)
-- SELECT
-- name,
--   COUNT(dur) AS count,
--   ROUND(SUM(dur) / 1e6, 2) as dur_ms,
--   ROUND(AVG(dur) /1e6, 4) AS avg_ms,
--   -- Calculate P50, P90, and P99 in
--   milliseconds
```

- Derived metrics
- Binding to a timestamp

CURRENT TRACE

cyclotron.pftrace (37 MB)

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Record new trace

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Open Chrome example

SETTINGS

Settings

Flags

Plugins

SUPPORT

Documentation & Bugs

CONVERT TRACE

Convert to other formats

Start: 06:03:34.652 108 800
Duration: 38m 7s 433ms 59µs 400ns

Default Workspace

Processes

Process 26724

GC Pause > 10ms

Delete Garbage > 10ms

Delete Garbage > 25%

Deduce Unreachable > 25%, > 10ms

Delete Garbage > 25%, > 10ms

Global Track Events

Process 26724 26724

Process 5072 5072

Process 7688 7688

Process 26472 26472

Process 16124 16124

Process 7460 7460

Process 1396 1396

Process 26840 26840

Process 26708 26708

Process 25680 25680

Process 14764 14764

Process 10788 10788

Process 25604 25604

Process 10732 10732

Process 7664 7664

Current Selection Query Page Results x Handle Resurrected (gen=2) (across trace) x

- Additional tracks for derived metrics

SUMMARY

- `gcmon` — an example of using the new API
- Lets you obtain additional information
- Uses Perfetto to store and visualize the data
- Perfetto can be used for your own purposes

BENCHMARKS

Problem ●

Tools ●

New API ●

gcmon ●

Perfetto ●

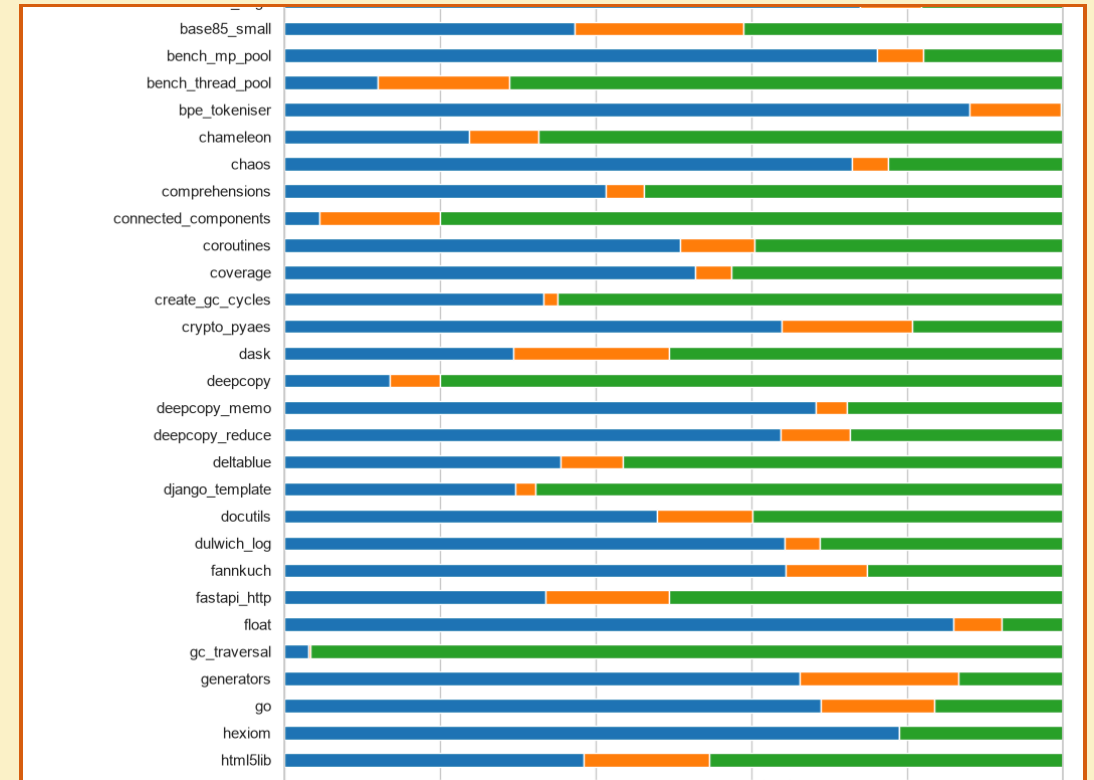
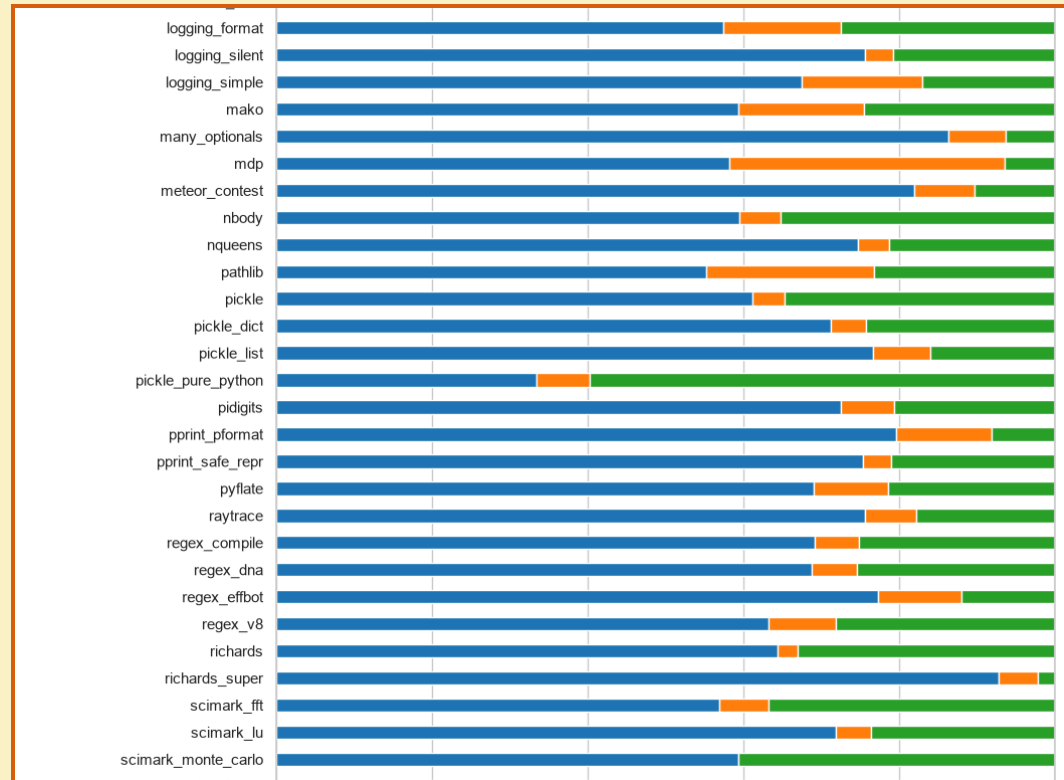
Benchmarks ●

Backports ○

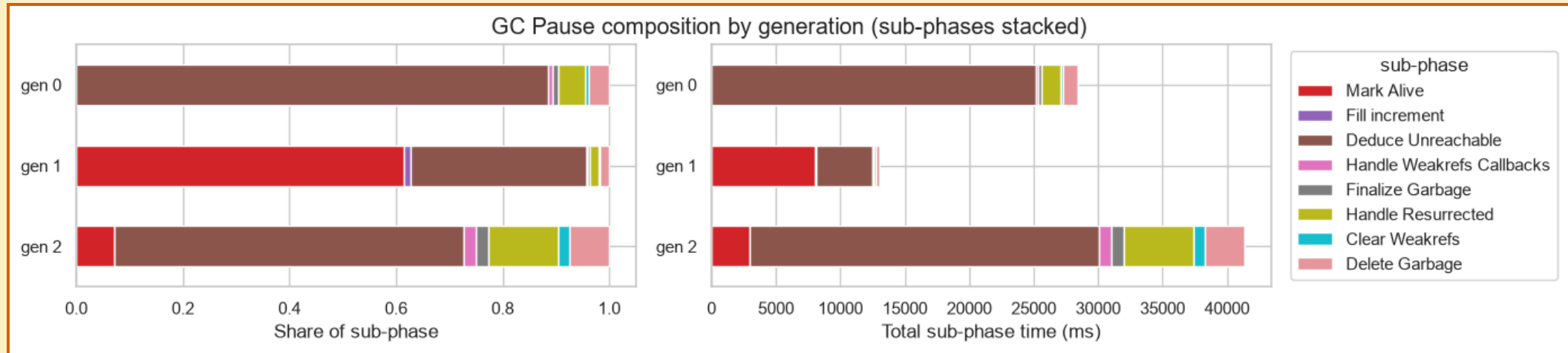
BENCHMARKS

- Why benchmarks?
- Pyperformance
- Cyclotron
- Any other

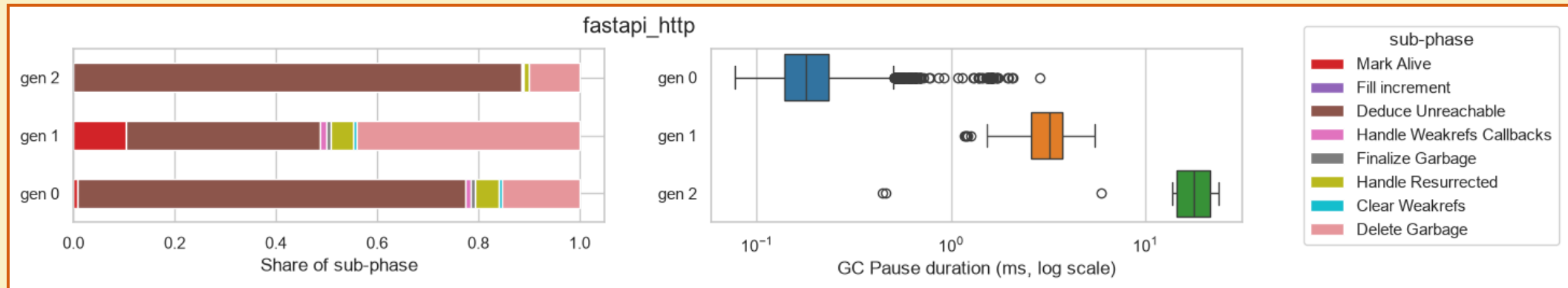
WHICH GENERATION DOMINATES?



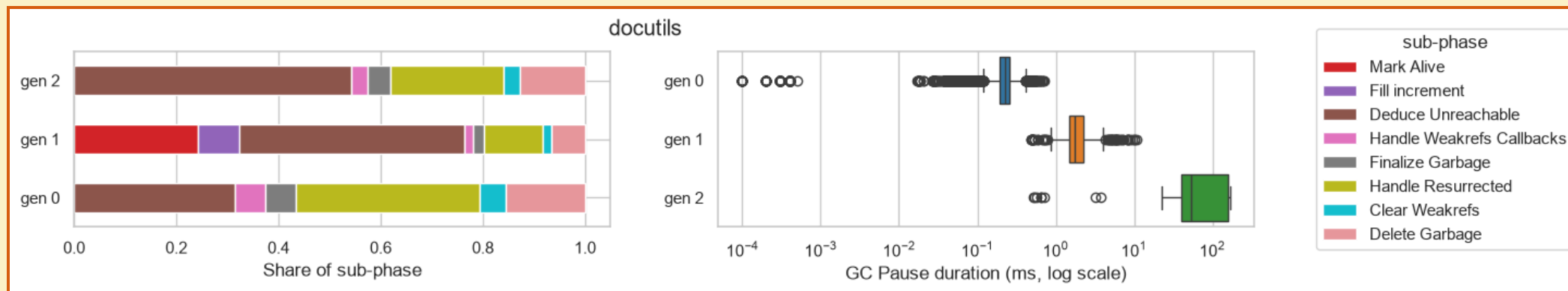
TYPICAL PHASE DISTRIBUTION



MANY OBJECTS WITH "HEAVY" DATA



FINALIZERS



SUMMARY

- `gcmon` / the new API
 - can be used to profile production systems
 - can be used to analyze garbage collector behavior
- You can obtain information about possible tuning of production systems

BACKPORTS

Problem ●

Tools ●

New API ●

gcmon ●

Perfetto ●

Benchmarks ●

Backports ●

3.13-3.14

```
typedef struct _Py_DebugOffsets {  
    char cookie[8] _Py_NONSTRING;  
    uint64_t version;  
    uint64_t free_threaded;  
    // Runtime state offset;  
    struct _runtime_state {  
        uint64_t size;  
        uint64_t finalizing;  
        uint64_t interpreters_head;  
    } runtime_state;  
    // Interpreter state offset;  
    struct _interpreter_state {  
        uint64_t size;  
        uint64_t id;  
        uint64_t next;  
        uint64_t gc;  
        // ...  
    } interpreter_state;  
    struct _gc {  
        uint64_t size;  
        uint64_t collecting;  
    } gc;  
} _Py_DebugOffsets;
```

```
/* Running stats per generation */  
struct gc_generation_stats {  
    Py_ssize_t collections;  
    Py_ssize_t collected;  
    Py_ssize_t uncollectable;  
};
```

```
struct _gc_runtime_state {  
    PyObject *trash_delete_later;  
    int trash_delete_nesting;  
    /* Is automatic collection enabled? */  
    int enabled;  
    int debug;  
    /* linked lists of container objects */  
    struct gc_generation generations[NUM_GENERATIONS];  
    PyGC_Head *generation0;  
    /* a permanent generation which won't be collected */  
    struct gc_generation permanent_generation;  
    struct gc_generation_stats generation_stats[NUM_GENERATIONS];  
    /* true if we are currently running the collector */  
    int collecting;  
    /* list of uncollectable objects */  
    PyObject *garbage;  
    /* a list of callbacks to be invoked when collection is performed */  
    PyObject *callbacks;  
};
```

3.13-3.14

```
gcscope tui --PID 4428 --Frame 220 @ 24.2s --Rate 100ms --Poll 57.8µs

_Py_DebugOffsets (embedded in _PyRuntime) @ 0x7ffbb98e3080 (size: 760 bytes)

Fields:
|_Py_DebugOffsets| Hex Dump (DebugOffsets, 0x0000-0x02f7, 760 bytes):
|--- 0x0000 cookie[s]| "xdebugpy"| 00000000 78 64 65 62 75 67 70 70 f0 01 0e 03 00 00 00 00 |xdebugpy.....
|--- 0x0008 version| 0x03e01f6| 00000010 00 00 00 00 00 00 00 00 a0 d0 04 00 00 00 00 00 |.....C.....
|--- 0x0010 free_threaded| 0| 00000020 10 03 00 00 00 00 00 00 20 13 00 00 00 00 00 00 |.....Q.....
|--- 0x0018 free_threaded| 0| 00000030 60 71 03 00 00 00 00 00 70 1c 00 00 00 00 00 00 |.....q.....
|--- runtime_state| 00000040 68 1c 00 00 00 00 00 00 a0 1c 00 00 00 00 00 00 |.....h.....
|--- 0x0018 size| 315560| 00000050 b8 1c 00 00 00 00 00 00 e0 1c 00 00 00 00 00 00 |.....
|--- 0x0020 finalizing| 784| 00000060 e0 1d 00 00 00 00 00 00 00 1d 00 00 00 00 00 00 |.....
|--- 0x0020 interpreters_head| 800| 00000070 e0 1d 00 00 00 00 00 00 10 00 00 00 00 00 00 00 |.....C.....
|--- interpreter_state| 00000080 40 1e 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |.....Q.....
|--- 0x0030 size| 225632| 00000090 50 1e 00 00 00 00 00 00 40 1e 00 00 00 00 00 00 |.....P.....H.....
|--- 0x0030 id| 7280| 000000a0 78 1e 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |.....X.....
|--- 0x0040 next| 7272| 000000b0 30 03 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |.....
|--- 0x0040 threads_head| 7336| 000000c0 00 00 00 00 00 00 00 00 10 00 00 00 00 00 00 00 |.....
|--- 0x0050 threads_main| 7352| 000000d0 40 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |.....H.....
|--- 0x0050 gc| 7400| 000000e0 9c 00 00 00 00 00 00 00 e0 00 00 00 00 00 00 00 |.....
|--- 0x0288 size| 240| 000000f0 20 00 00 00 00 00 00 00 50 00 00 00 00 00 00 00 |.....X.....
|--- 0x0290 collecting| 192| 00000100 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |.....
|--- 0x0290 collecting| 192| 00000110 30 00 00 00 00 00 00 00 50 00 00 00 00 00 00 00 |.....B.....
|--- 0x0290 collecting| 192| 00000120 40 00 00 00 00 00 00 00 40 00 00 00 00 00 00 00 |.....J.....
|--- 0x0290 collecting| 192| 00000130 00 00 00 00 00 00 00 00 d0 00 00 00 00 00 00 00 |.....
|--- 0x0290 collecting| 192| 00000140 70 00 00 00 00 00 00 00 70 00 00 00 00 00 00 00 |.....p.....X.....
|--- 0x0290 collecting| 192| 00000150 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |.....
|--- 0x0290 collecting| 192| 00000160 44 00 00 00 00 00 00 00 34 00 00 00 00 00 00 00 |.....D.....
|--- 0x0290 collecting| 192| 00000170 60 00 00 00 00 00 00 00 60 00 00 00 00 00 00 00 |.....
|--- 0x0290 collecting| 192| 00000180 d0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |.....
|--- 0x0290 collecting| 192| 00000190 10 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |.....
|--- 0x0290 collecting| 192| 000001a0 a0 01 00 00 00 00 00 10 00 00 00 00 00 00 00 |.....
|--- 0x0290 collecting| 192| 000001b0 50 00 00 00 00 00 00 00 a0 00 00 00 00 00 00 00 |.....X.....
|--- 0x0290 collecting| 192| 000001c0 20 00 00 00 00 00 00 00 20 00 00 00 00 00 00 00 |.....C.....
|--- 0x0290 collecting| 192| 000001d0 10 00 00 00 00 00 00 00 20 00 00 00 00 00 00 00 |.....
|--- 0x0290 collecting| 192| 000001e0 10 00 00 00 00 00 00 00 10 00 00 00 00 00 00 00 |.....
|--- 0x0290 collecting| 192| 000001f0 c0 00 00 00 00 00 00 00 10 00 00 00 00 00 00 00 |.....
|--- 0x0290 collecting| 192| 00000200 20 00 00 00 00 00 00 00 20 00 00 00 00 00 00 00 |.....C.....
|--- 0x0290 collecting| 192| 00000210 30 00 00 00 00 00 00 00 20 00 00 00 00 00 00 00 |.....B.....
|--- 0x0290 collecting| 192| 00000220 20 00 00 00 00 00 00 00 10 00 00 00 00 00 00 00 |.....C.....
|--- 0x0290 collecting| 192| 00000230 10 00 00 00 00 00 00 00 20 00 00 00 00 00 00 00 |.....
|--- 0x0290 collecting| 192| 00000240 10 00 00 00 00 00 00 00 10 00 00 00 00 00 00 00 |.....
|--- 0x0290 collecting| 192| 00000250 20 00 00 00 00 00 00 00 10 00 00 00 00 00 00 00 |.....C.....
|--- 0x0290 collecting| 192| 00000260 20 00 00 00 00 00 00 00 40 00 00 00 00 00 00 00 |.....Q.....
|--- 0x0290 collecting| 192| 00000270 20 00 00 00 00 00 00 00 10 00 00 00 00 00 00 00 |.....
|--- 0x0290 collecting| 192| 00000280 20 00 00 00 00 00 00 00 f0 00 00 00 00 00 00 00 |.....C.....
|--- 0x0290 collecting| 192| 00000290 c0 00 00 00 00 00 00 00 a0 00 00 00 00 00 00 00 |.....
|--- 0x0290 collecting| 192| 000002a0 10 00 00 00 00 00 00 00 40 00 00 00 00 00 00 00 |.....H.....
|--- 0x0290 collecting| 192| 000002b0 43 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |.....C.....
|--- 0x0290 collecting| 192| 000002c0 00 00 00 00 00 00 00 00 10 00 00 00 00 00 00 00 |.....
|--- 0x0290 collecting| 192| 000002d0 20 01 00 00 00 00 00 00 c0 1e 00 00 00 00 00 00 |.....C.....
|--- 0x0290 collecting| 192| 000002e0 00 00 00 00 00 00 00 00 04 00 00 00 00 00 00 00 |.....
|--- 0x0290 collecting| 192| 000002f0 00 02 00 00 00 00 00 00 |.....

PyInterpreterState @ 0x7ffbb98f0f40 (struct: 225632 bytes)

Key offset values stored in _Py_DebugOffsets:
|key offset values stored in _Py_DebugOffsets:| GC struct (240 bytes) hex dump:
|--- 0x0000| 00 00 00 00 00 00 00 00 00 00 00 00 01 00 00 00 |.....
|--- 0x0010| 00 00 00 00 00 00 00 00 60 e0 24 00 0d 02 00 00 |.....$.
|--- 0x0020| a0 8f a7 00 0d 02 00 00 d0 07 00 00 c4 07 00 00 |.....
|--- 0x0030| e0 e7 51 00 0d 02 00 00 b0 ad 31 00 0d 02 00 00 |.....Q.....
|--- 0x0040| 00 00 00 00 00 00 00 00 b0 d2 2b 00 0d 02 00 00 |.....J.....
|--- 0x0050| 00 00 00 00 00 00 00 00 90 ad 8f b9 fb 00 00 00 |.....
|--- 0x0060| 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |.....
|--- 0x0070| 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |.....
|--- 0x0080| 00 00 00 00 00 00 00 00 f4 d0 2c 4e 00 00 00 00 |.....N.....
```

_Py_DebugOffsets

```
gcscope tui --PID 4428 --Frame 522 @ 57.7s --Rate 100ms --Poll 60.4µs

GC Stats Buffer View ([g] back to full layout)

Buffer @ 0x7ffbb98fac08 (size: 72 bytes, entry: 24 bytes)
|Gen 0 (Young) - 1 entries (rate = n/a, avg coll = n/a)| 00000000 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |.....
|Gen 1 (Middle) - 1 entries (rate = n/a, avg coll = n/a)| 00000010 00 00 00 00 00 00 00 00 d2 b5 03 00 00 00 00 00 |.....
|Gen 2 (Oldest) - 1 entries (rate = n/a, avg coll = n/a)| 00000020 50 ea 30 4e 00 00 00 00 00 00 00 00 00 00 00 00 |.....
|00000030 01 00 00 00 00 00 00 00 15 00 00 00 00 00 00 00 |.....
|00000040 00 00 00 00 00 00 00 00 |.....

gen idx collections collected heap duration(s)
|--- 0 0 0 0 0 0.000|
|1 0 243154 1311020574 0 0.000|
|2 0 1 21 0 0.000|

Entry #1 (gen 0, entry 0) @ 0x7ffbb98fac08
|collections @ +0 0|
|collected @ +0 0|
|uncollectable @ +16 0|
```

GC Stats Buffer

3.13-3.14

```
gcscope tui —PID 4428 —Frame 522 @ 57.7s —Rate 188ms —Poll 68.4μs
GC Stats Buffer View ([g] back to full layout)

Buffer @ 0x7ffb98faca8 (size: 72 bytes, entry: 24 bytes)
|Gen 0 (Young) - 1 entries (rate = n/a, avg coll = n/a)
|Gen 1 (Middle) - 1 entries (rate = n/a, avg coll = n/a)
|Gen 2 (Oldest) - 1 entries (rate = n/a, avg coll = n/a)
|.....|
|gen  idx  collections  collected  heap  duration(s)|
|.....|
| 0      0              0            0          0      0.888|
| 1      0      243154    1311828574    0      0.888|
| 2      0              1            21          0      0.888|
|.....|
Entry #1 (gen 0, entry 0) @ 0x7ffb98faca8
|collections @ +0  0|
|collected @ +0  0|
|uncollectable @ +16 0|
|.....|

ts_start @ +0 266.436.533.253.680
ts_stop @ +0 266.436.539.872.680
collections @ +16 594869
collected @ +24 1186154611
uncollectable @ +32 0
candidates @ +40 1191522914
duration @ +48 3672.386371
heap_size @ +56 11778
```

3.14

```
gcscope tui —PID 24288 —Frame 766 @ 83.1s —Rate 188ms —Poll 76.1μs
GC Stats Buffer View ([g] back to full layout)

Buffer @ 0x198d51de470 (size: 1112 bytes, entry: 64 bytes)
|Gen 0 (Young) - 11 entries (rate = 3.93/s, avg coll = 6.888ms)
|Gen 1 (Middle) - 3 entries (rate = 0.32/s, avg coll = 5.632ms)
|Gen 2 (Oldest) - 3 entries (rate = 0.8/s, avg coll = 10.781ms)
|.....|
|gen  idx  collections  collected  heap  duration(s)|
|.....|
| 0      0      594869    1186154611    11.8K    3672.386|
| 0      1      594870    1186156605    11.8K    3672.393|
| 0      2      594871    1186158599    11.8K    3672.399|
| 0      3      594872    1186160593    11.8K    3672.405|
| 0      4      594862    1186148653    11.8K    3672.344|
| 0      5      594865    1186142647    11.8K    3672.351|
| 0      6      594864    1186144641    11.8K    3672.357|
| 0      7      594865    1186146635    11.8K    3672.363|
| 0      8      594866    1186148629    11.8K    3672.369|
| 0      9      594867    1186158623    11.8K    3672.375|
| 0     10      594868    1186152617    11.8K    3672.381|
| 1      0      54870     107831532    11.8K    334.103|
| 1      1      54876     107827544    11.8K    334.092|
| 1      2      54877     107829538    11.8K    334.098|
| 2      0        156      389878    11.8K    1.273|
| 2      1        154      385882    11.8K    1.252|
| 2      2        155      387876    11.8K    1.268|
|.....|
Entry #1 (gen 0, entry 0) @ 0x198d51de470
|ts_start @ +0 266.436.533.253.680|
|ts_stop @ +0 266.436.539.872.680|
|collections @ +16 594869|
|collected @ +24 1186154611|
|uncollectable @ +32 0|
|candidates @ +40 1191522914|
|duration @ +48 3672.386371|
|heap_size @ +56 11778|
```

3.15+

3.8, 3.9-3.10, 3.11-3.12

- What if there's no marker, nothing at all — what do we do?
- How stable is it?
- What factors can affect it?
- How do we validate it?



The screenshot displays the gcscope GUI for PID 27784, Frame 549, at 59.9s. The interface is divided into several sections:

- GC Stats Buffer View:** Shows a table of GC statistics for generations 0 (Young), 1 (Middle), and 2 (Oldest). Each generation has 1 entry.
- Buffer View:** Displays a memory buffer at address 0x7ffb5e98378, size 72 bytes, with 24 entries. It shows a hex dump and ASCII representation of the buffer contents.
- Entry #1:** Provides details for the first entry of generation 0, including its index (0), collections (0), collected value (2435222), heap duration (0.000s), and uncollectable value (0).

An orange box with the number "3.8" is overlaid on the bottom left of the screenshot.

CALL TO ACTION

- Which metrics are interesting and/or necessary for you?
- In what scenarios would you use these capabilities?
- Which formats and integrations interest you?
- What is your environment?
- What alerts do you envision?

THANK YOU FOR YOUR ATTENTION!

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